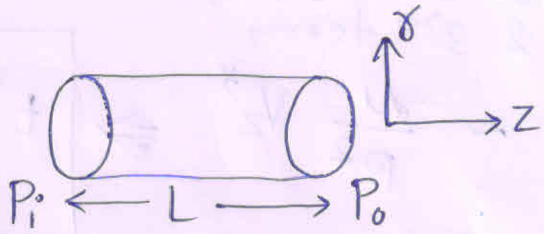

2 marks

(a) governing eqn :-

$$\boxed{\rho \frac{\partial V_z}{\partial t} = -\frac{\partial P}{\partial Z} + \mu \frac{1}{r}\left(\frac{\partial}{\partial r}\left(r \frac{\partial V_z}{\partial r}\right)\right)}$$

$$\frac{\partial P}{\partial Z} = \frac{P_o - P_i}{L}$$ (P is decreasing a fun of Z)

(2)

I.C. $V_z(t=0, r) = 0$

B.C. $V_z(r=R, \forall t) = 0$ (no slip)

$V_z(r=0, \forall t) =$ finite

or $\left.\frac{\partial V_z}{\partial r}\right|_{r=0} = 0$

from symmetry

(b) order of magnitude analysis,

(1) mark

$$\underbrace{\rho \frac{V_z^*}{t^*} \sim \frac{P_i - P_o}{L} \sim \frac{\mu}{R}\left(\frac{1}{R}\left(\frac{R V_z^*}{R}\right)\right)}_{\text{all must have same order of magnitude.}}$$

1st term 2nd term 3rd term

Considering 1st and 3rd term,

Considering 1st & 3rd term

$$\rho \frac{V_z^*}{t^*} \sim \frac{\mu}{R^2} V_z^* \Rightarrow \boxed{t^* \sim \frac{R^2}{\mu/\rho}}$$ ① mark

Considering 2nd and 3rd term,

① mark

$$\frac{P_i - P_o}{L} \sim \frac{\mu}{R^2} V_z^* \Rightarrow \boxed{V_z^* \sim \frac{(P_i - P_o) R^2}{\mu L}}$$

ⓒ $\eta = \frac{r}{R}$, $\tau = \frac{t}{t^*}$, $\phi = \frac{V_z}{V_z^*}$

if $\frac{\partial P}{\partial z}$ is linear, we can directly use $P_i - P_o$ & L as characterstics values

$$\rho \frac{\partial V_z}{\partial t} = -\frac{\partial P}{\partial z} + \frac{\mu}{r}\left(\frac{\partial}{\partial r}\left(r \frac{\partial V_z}{\partial r}\right)\right)$$

$$\frac{V_z^*}{t^*} \rho \frac{\partial \phi}{\partial \tau} = (-)\frac{P_o - P_i}{L} + \frac{\mu}{R^2 \eta} \frac{\partial}{\partial \eta}\left(\frac{\eta \cancel{R}}{\cancel{R}} V_z^* \frac{\partial \phi}{\partial \eta}\right)$$

$$\rho \frac{V_z^*}{t^*} \frac{\partial \phi}{\partial \tau} = \frac{P_i - P_o}{L} + \frac{\mu V_z^*}{R^2 \eta} \frac{\partial}{\partial \eta}\left(\eta \frac{\partial \phi}{\partial \eta}\right)$$

$$\frac{\rho}{t^*} \frac{\partial \phi}{\partial \tau} = \frac{(P_i - P_o)}{V_z^* L} + \frac{\mu}{R^2 \eta} \frac{\partial}{\partial \eta}\left(\eta \frac{\partial \phi}{\partial \eta}\right)$$

Using values of V_z^*, t^*,

$$\frac{\cancel{\rho}}{R^2(\mu/\cancel{\rho})} \frac{\partial \phi}{\partial \tau} = \frac{\frac{(P_i - \cancel{P_o})}{(P_i \cancel{-} P_o) R^2 \cancel{L}}}{4 \mu \cancel{L}} + \frac{\mu}{R^2 \eta} \frac{\partial}{\partial \eta}\left(\eta \frac{\partial \phi}{\partial \eta}\right)$$

[I have proportionally taken $v_z^* = \frac{(P_i - P_o) R^2}{4 \mu L}$ to get rid of factor 4,

$$\frac{\cancel{\mu}}{\cancel{R^2}} \frac{\partial \phi}{\partial \tau} = \frac{4 \cancel{\mu}}{\cancel{R^2}} + \frac{\cancel{\mu}}{\cancel{R^2}} \frac{\partial}{\eta \, \partial \eta}\left(\eta \frac{\partial \phi}{\partial \eta}\right)$$

(however even if you take $V_z^* = \frac{(P_i - P_o) R^2}{\mu L}$, you will get the same Answer.)

$$\boxed{\frac{\partial \phi}{\partial \tau} = 4 + \frac{1}{\eta} \frac{\partial}{\partial \eta}\left(\eta \frac{\partial \phi}{\partial \eta}\right)}$$

2 marks only if you will get final form correctly.

d $\phi = \phi_s + \phi_t$

we have $\cancel{\frac{\partial \phi}{\partial \tau}} = 4 + \frac{1}{\eta} \frac{\partial}{\partial \eta}\left(\eta \frac{\partial \phi}{\partial \eta}\right)$

$$\Rightarrow \frac{\partial}{\partial \eta}\left(\eta \frac{\partial \phi}{\partial \eta}\right) = -4\eta \Rightarrow$$

$$\eta \frac{\partial \phi}{\partial \eta} = -\frac{4\eta^2}{2} + C$$

$$\Rightarrow \frac{\partial \phi}{\partial \eta} = -2\eta + \frac{C}{\eta}$$

$$\phi = -\eta^2 + c \ln \eta + C_1$$

B.C. ① $r \to 0$, V_z finite

$\Rightarrow \eta \to 0$, ϕ is finite

$\Rightarrow c = 0 \quad \because (\ln(\eta \to 0) \to -\infty)$

here $\phi = -\eta^2 + C_1$

B.C. ② $\underbrace{\eta = 1, \ \phi = 0}_{(r=R) \quad V_z = 0} \Rightarrow C_1 = 1$

$$\boxed{\phi_{ss} = 1 - \eta^2}$$ Steady state solution — ③ marks

ⓔ governing eq^n (a)

$$\frac{\partial \phi}{\partial \tau} = 4 + \frac{1}{\eta} \frac{\partial}{\partial \eta}\left(\eta \frac{\partial \phi}{\partial \eta}\right)$$

$$\frac{\partial(\phi_{ss} + \phi_t)}{\partial \tau} = 4 + \frac{1}{\eta} \frac{\partial}{\partial \eta}\left(\eta \frac{\partial(\phi_s + \phi_t)}{\partial \eta}\right)$$

$$\cancelto{0}{\frac{\partial \phi_{ss}}{\partial \tau}} + \frac{\partial \phi_t}{\partial \tau} = 4 + \frac{1}{\eta} \frac{\partial}{\partial \eta}\left(\eta \cdot \left(\frac{\partial \phi_{ss}}{\partial \eta} + \frac{\partial \phi_t}{\partial \eta}\right)\right)$$

$$\frac{\partial \phi_t}{\partial \tau} = 4 + \frac{1}{\eta} \frac{\partial}{\partial \eta}\left(\eta\left(-2\eta + \frac{\partial \phi_t}{\partial \eta}\right)\right)$$

$$\frac{\partial \phi_t}{\partial \tau} = 4 + \frac{1}{\eta}\left[(-4\eta) + \frac{\partial}{\partial \eta}\left(\eta \frac{\partial \phi_t}{\partial \eta}\right)\right]$$

$$\frac{\partial \phi_t}{\partial \tau} = \cancel{4} + (\cancel{-4}) + \frac{1}{\eta}\frac{\partial}{\partial \eta}\left(\eta \frac{\partial \phi_t}{\partial \eta}\right)$$

$$\boxed{\frac{\partial \phi_t}{\partial \tau} = \frac{1}{\eta}\left(\frac{\partial}{\partial \eta}\left(\eta \frac{\partial \phi_t}{\partial \eta}\right)\right)}$$ — 3 marks

(f) let us solve $\frac{\partial \phi_t}{\partial \tau} = \frac{1}{\eta}\frac{\partial}{\partial \eta}\left(\eta \frac{\partial \phi_t}{\partial \eta}\right)$ — eqn. (b)

I.C. $\phi(\eta, t=0) = 0$

$\phi_{ss} + \phi_t(t, \eta = 0) = 0$

$\phi_t(\eta, t=0) = -\phi_{ss}(\eta)$

$= \eta^2 - 1$

$\xRightarrow{I.C.} \phi_t(\eta, \tau=0) = \eta^2-1$ (1)

B.C. overall B.C. & steady state B.C. must be identical,

Hence, $\phi_t(\eta=1, \forall \tau) = 0$

$\phi_t(\eta=0, \forall \tau) = \text{finite}$ (1)

let us assume/ propose,

$$\phi_t = f(\eta)\, g(\tau)$$

putting it in eq^n (b)

$$f \frac{\partial g}{\partial \tau} = \frac{1}{\eta}\frac{\partial}{\partial \eta}\left(\eta\, g \frac{\partial f}{\partial \eta}\right)$$

$$\frac{1}{g}\frac{\partial g}{\partial \tau} = \frac{1}{\eta f}\frac{\partial\left(\eta \frac{\partial f}{\partial \eta}\right)}{\partial \eta} = -c^2$$

$$\frac{1}{g}\frac{dg}{d\tau} = -c^2 \Rightarrow \quad g = De^{-c^2\tau}$$

$$\frac{1}{\eta f}\frac{\partial\left(\eta \frac{\partial f}{\partial \eta}\right)}{\partial \eta} = -c^2$$

rearrangine.

$$\frac{d^2 f}{d\eta^2} + \frac{1}{\eta}\frac{\partial f}{\partial \eta} + c^2 f = 0$$

$\Rightarrow$ rearranging

$$\eta^2 c^2 \frac{d^2 f}{d(\eta^2 c^2)} + \eta c \frac{\partial f}{\partial \eta c} + c^2\eta^2 f = 0$$

$$\Rightarrow (\eta c)^2 \frac{d^2 f}{d(\eta c)^2} + (\eta c)\frac{\partial f}{\partial(\eta c)} + (c\eta)^2 f = 0$$

Let say $\eta c = q$

$$\Rightarrow q^2 \frac{\partial^2 f}{\partial q^2} + q\frac{\partial f}{\partial q} + (q^2 - 0^2) f = 0$$

(comparing it with standard Bessel diff. eqn.)

we get 0^{th} order solution

$$\Rightarrow f = AJ_0(q) + BY_0(q)$$

$$f = AJ_0(c\eta) + BY_0(c\eta) \Rightarrow \frac{\partial f}{\partial \eta} = -AJ_1(c\eta) + (-BY_1(c\eta))$$

$$= -AJ_1(c\eta) - BY_1(c\eta)$$

applying B.C. $\eta = 0$ $(r = 0)$ $\frac{\partial f}{\partial \eta} = 0$ $\Rightarrow$ $-AJ_1(0) - BY_1(0) = 0$

$$-A(0) - B(-\infty) = 0$$

$$\boxed{0 = B} \Leftarrow 0 - B(-\infty) = 0$$

Hence $f = A J_0(c\eta)$

applying another B.C. $\eta = 1$; $f = 0$
$(r = R)$

$$\Rightarrow 0 = A J_0(c) \Rightarrow J_0(c) = 0 \quad \text{— ①}$$

$$c_n = c_1, c_2 \ldots\ldots c_\infty$$

(roots of eqn ①)

[you can check on the internet value of c.
$c_1 = 2.4048$
$c_2 = 5.5201$
$c_n = \cdots$]

Hence the general solution we have

$$\phi = \left(\sum_{n=1}^{\infty} D_n e^{-c_n^2 \tau} A_n J_0(c_n \eta) \right) + \phi_{ss}$$

$$= \sum_{n=1}^{\infty} A_n D_n J_0(c_n \eta) e^{-c_n^2 \tau} + (1 - \eta^2)$$

$$\phi = \sum_{n=1}^{\infty} P_n J_0(c_n \eta) e^{-c_n^2 \tau} + (1 - \eta^2)$$

Let apply initial condition, $\tau = 0$, $\phi = 0$

$$\Rightarrow \boxed{\eta^2 - 1 = \sum_{n=1}^{\infty} P_n J_0(c_n \eta)}$$

multiplying both sides by $\eta J_0(c_m \eta)$
and integrating w.r.t. η.

$$\Rightarrow \int_0^1 \eta(\eta^2 - 1) J_0(c_m \eta) d\eta = \int_0^1 \sum_{n=1}^{\infty} \left(P_n J_0(c_n \eta) \right) \eta J_0(c_m \eta) d\eta$$

using orthogonality condition
(check on internet)

$$\int_0^1 \eta J_0(c_n \eta) J_0(c_m \eta) d\eta = 0$$

$$(-)\int_0^1 \eta\,(J_0(c_m\eta))(1-\eta^2)\,d\eta = \sum_{\eta=1}^{\infty} Q_n \int_0^1 \eta\, J_0(c_m\eta)\,J_0(c_m\eta)\,d\eta$$

$$(-)\int_0^1 \eta(1-\eta^2)\,J_0(c_m\eta)\,d\eta = \sum_{n=1}^{\infty} Q_n \int_0^1 \eta\, J_0(c_m\eta)\,J_0(c_n\eta)\,d\eta$$

$m \neq n$ term on right hand side drops (orthogonality)

hence $m=n$ (only term survives $m=n$)

$$(-)\int_0^1 \eta(1-\eta^2)\,J_0(c_n\eta)\,d\eta = Q_n \underbrace{\int_0^1 \eta\,(J_0(c_n\eta))^2\,d\eta}_{\phi \text{ given}}$$

$$(-)\,\frac{4\,(J_1(c_n))}{c_n^3} = Q_n\,\frac{1}{2}\,(J_1(c_n))^2$$

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$$\frac{-8}{c_n^3\,J_1(c_n)} = Q_n$$

Hence, $\phi = \phi_{ss} + \phi_1 = Q_n(1-\eta^2) - \dfrac{8}{c_n^3 J_1(c_n)}\,J_0(c_n\eta)\,\exp(-c_n^2\tau)$

$$\boxed{\phi = (1-\eta^2) - \frac{8}{c_n^3\,J_1(c_n)}\,J_0(c_n\eta)\,\exp(-c_n^2\tau)}$$